

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–13

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

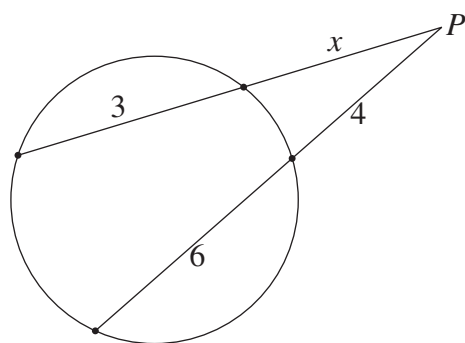
1 What is the remainder when $x^3 - 6x$ is divided by $x + 3$?

- (A) -9
- (B) 9
- (C) $x^2 - 2x$
- (D) $x^2 - 3x + 3$

2 Given that $N = 100 + 80e^{kt}$, which expression is equal to $\frac{dN}{dt}$?

- (A) $k(100 - N)$
- (B) $k(180 - N)$
- (C) $k(N - 100)$
- (D) $k(N - 180)$

3 Two secants from the point P intersect a circle as shown in the diagram.



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SCALE

What is the value of x ?

- (A) 2
- (B) 5
- (C) 7
- (D) 8

- 4 A rowing team consists of 8 rowers and a coxswain.

The rowers are selected from 12 students in Year 10.

The coxswain is selected from 4 students in Year 9.

In how many ways could the team be selected?

(A) ${}^{12}C_8 + {}^4C_1$

(B) ${}^{12}P_8 + {}^4P_1$

(C) ${}^{12}C_8 \times {}^4C_1$

(D) ${}^{12}P_8 \times {}^4P_1$

- 5 What are the asymptotes of $y = \frac{3x}{(x+1)(x+2)}$?

(A) $y = 0, \quad x = -1, \quad x = -2$

(B) $y = 0, \quad x = 1, \quad x = 2$

(C) $y = 3, \quad x = -1, \quad x = -2$

(D) $y = 3, \quad x = 1, \quad x = 2$

- 6 What is the domain of the function $f(x) = \sin^{-1}(2x)$?

(A) $-\pi \leq x \leq \pi$

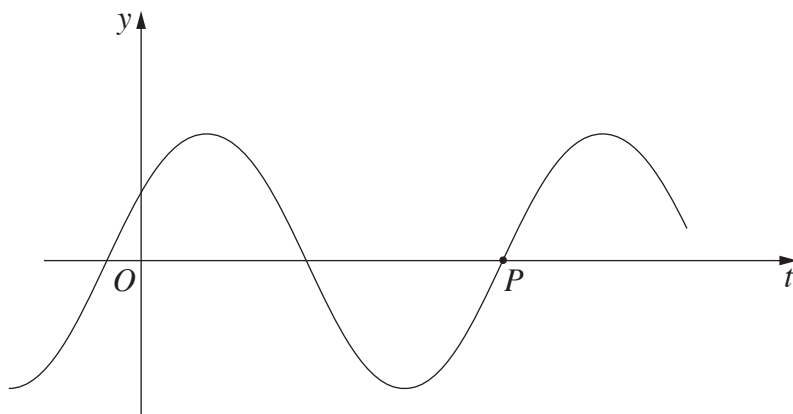
(B) $-2 \leq x \leq 2$

(C) $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

(D) $-\frac{1}{2} \leq x \leq \frac{1}{2}$

- 7 What is the value of k such that $\int_0^k \frac{1}{\sqrt{4-x^2}} dx = \frac{\pi}{3}$?
- (A) 1
(B) $\sqrt{3}$
(C) 2
(D) $2\sqrt{3}$
- 8 What is the value of $\lim_{x \rightarrow 3} \frac{\sin(x-3)}{(x-3)(x+2)}$?
- (A) 0
(B) $\frac{1}{5}$
(C) 5
(D) Undefined
- 9 Two particles oscillate horizontally. The displacement of the first is given by $x = 3 \sin 4t$ and the displacement of the second is given by $x = a \sin nt$. In one oscillation, the second particle covers twice the distance of the first particle, but in half the time.
- What are the values of a and n ?
- (A) $a = 1.5, n = 2$
(B) $a = 1.5, n = 8$
(C) $a = 6, n = 2$
(D) $a = 6, n = 8$

- 10 The graph of the function $y = \cos\left(2t - \frac{\pi}{3}\right)$ is shown below.



What are the coordinates of the point P ?

- (A) $\left(\frac{5\pi}{12}, 0\right)$
- (B) $\left(\frac{2\pi}{3}, 0\right)$
- (C) $\left(\frac{11\pi}{12}, 0\right)$
- (D) $\left(\frac{7\pi}{6}, 0\right)$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int \sin^2 x \, dx$. **2**

(b) Calculate the size of the acute angle between the lines $y = 2x + 5$ and $y = 4 - 3x$. **2**

(c) Solve the inequality $\frac{4}{x+3} \geq 1$. **3**

(d) Express $5 \cos x - 12 \sin x$ in the form $A \cos(x + \alpha)$, where $0 \leq \alpha \leq \frac{\pi}{2}$. **2**

(e) Use the substitution $u = 2x - 1$ to evaluate $\int_1^2 \frac{x}{(2x-1)^2} \, dx$. **3**

(f) Consider the polynomials $P(x) = x^3 - kx^2 + 5x + 12$ and $A(x) = x - 3$.

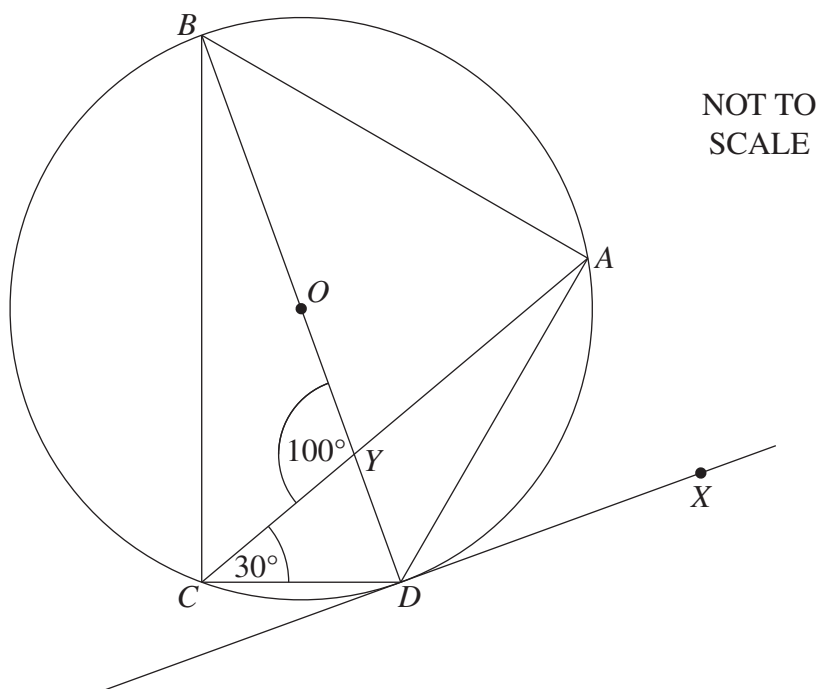
(i) Given that $P(x)$ is divisible by $A(x)$, show that $k = 6$. **1**

(ii) Find all the zeros of $P(x)$ when $k = 6$. **2**

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) In the diagram, the points A , B , C and D are on the circumference of a circle, whose centre O lies on BD . The chord AC intersects the diameter BD at Y . The tangent at D passes through the point X .

It is given that $\angle CYB = 100^\circ$ and $\angle DCY = 30^\circ$.



Copy or trace the diagram into your writing booklet.

- | | |
|--|----------|
| (i) What is the size of $\angle ACB$? | 1 |
| (ii) What is the size of $\angle ADX$? | 1 |
| (iii) Find, giving reasons, the size of $\angle CAB$. | 2 |

Question 12 continues on page 8

Question 12 (continued)

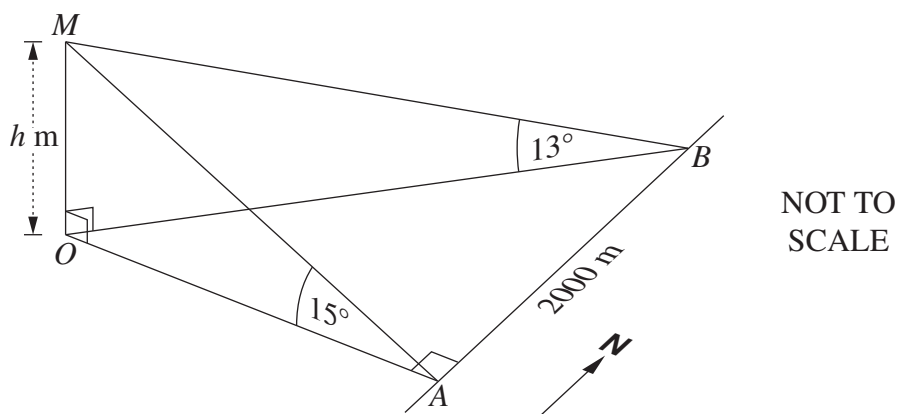
- (b) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

The equation of the chord PQ is given by $(p + q)x - 2y - 2apq = 0$. (Do NOT prove this.)

- (i) Show that if PQ is a focal chord then $pq = -1$. 1
 - (ii) If PQ is a focal chord and P has coordinates $(8a, 16a)$, what are the coordinates of Q in terms of a ? 2
- (c) A person walks 2000 metres due north along a road from point A to point B . The point A is due east of a mountain OM , where M is the top of the mountain. The point O is directly below point M and is on the same horizontal plane as the road. The height of the mountain above point O is h metres.

From point A , the angle of elevation to the top of the mountain is 15° .

From point B , the angle of elevation to the top of the mountain is 13° .

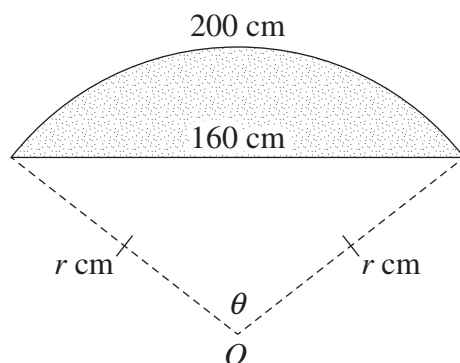


- (i) Show that $OA = h \cot 15^\circ$. 1
- (ii) Hence, find the value of h . 2

Question 12 continues on page 9

Question 12 (continued)

- (d) A kitchen bench is in the shape of a segment of a circle. The segment is bounded by an arc of length 200 cm and a chord of length 160 cm. The radius of the circle is r cm and the chord subtends an angle θ at the centre O of the circle.



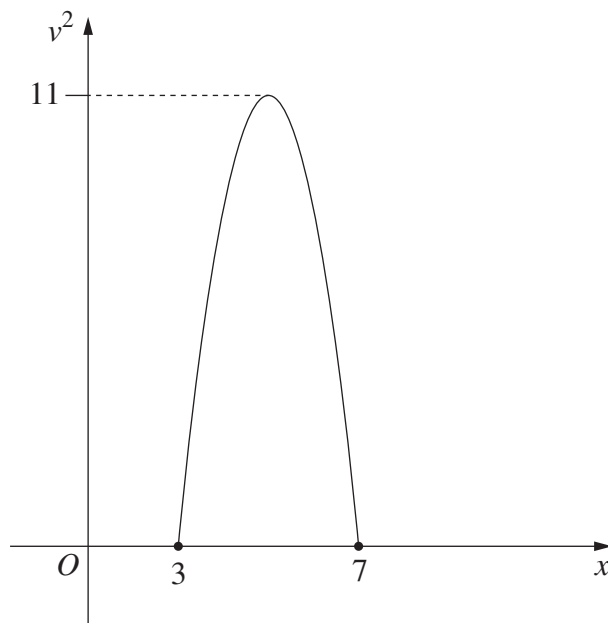
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- (i) Show that $160^2 = 2r^2(1 - \cos \theta)$. 1
- (ii) Hence, or otherwise, show that $8\theta^2 + 25 \cos \theta - 25 = 0$. 2
- (iii) Taking $\theta_1 = \pi$ as a first approximation to the value of θ , use one application of Newton's method to find a second approximation to the value of θ . Give your answer correct to two decimal places. 2

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) A particle is moving along the x -axis in simple harmonic motion. The displacement of the particle is x metres and its velocity is $v \text{ m s}^{-1}$. The parabola below shows v^2 as a function of x .



- | | | |
|-------|---|----------|
| (i) | For what value(s) of x is the particle at rest? | 1 |
| (ii) | What is the maximum speed of the particle? | 1 |
| (iii) | The velocity v of the particle is given by the equation | 3 |

$$v^2 = n^2(a^2 - (x - c)^2)$$

where a , c and n are positive constants.

What are the values of a , c and n ?

Question 13 continues on page 11

Question 13 (continued)

- (b) Consider the binomial expansion

$$\left(2x + \frac{1}{3x}\right)^{18} = a_0x^{18} + a_1x^{16} + a_2x^{14} + \dots$$

where $a_0, a_1, a_2, \dots$ are constants.

- (i) Find an expression for a_2 . **2**
- (ii) Find an expression for the term independent of x . **2**
- (c) Prove by mathematical induction that for all integers $n \geq 1$, **3**

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}.$$

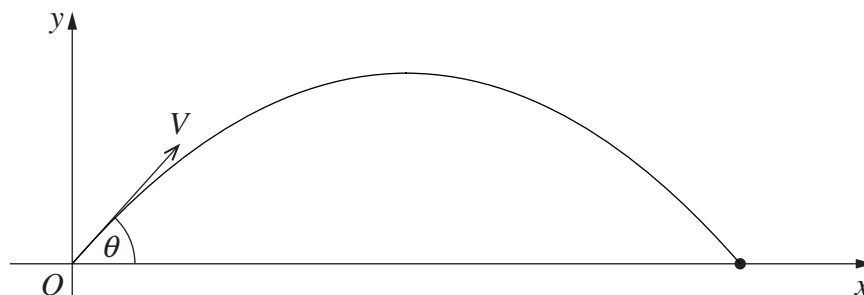
- (d) Let $f(x) = \cos^{-1}(x) + \cos^{-1}(-x)$, where $-1 \leq x \leq 1$.
- (i) By considering the derivative of $f(x)$, prove that $f(x)$ is constant. **2**
- (ii) Hence deduce that $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$. **1**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) A projectile is fired from the origin O with initial velocity $V \text{ m s}^{-1}$ at an angle θ to the horizontal. The equations of motion are given by

$$x = Vt \cos \theta, \quad y = Vt \sin \theta - \frac{1}{2}gt^2. \quad (\text{Do NOT prove this.})$$



- (i) Show that the horizontal range of the projectile is $\frac{V^2 \sin 2\theta}{g}$. **2**

A particular projectile is fired so that $\theta = \frac{\pi}{3}$.

- (ii) Find the angle that this projectile makes with the horizontal when **2**
 $t = \frac{2V}{\sqrt{3}g}$.

- (iii) State whether this projectile is travelling upwards or downwards when **1**
 $t = \frac{2V}{\sqrt{3}g}$. Justify your answer.

Question 14 continues on page 13

Question 14 (continued)

- (b) A particle is moving horizontally. Initially the particle is at the origin O moving with velocity 1 m s^{-1} .

The acceleration of the particle is given by $\ddot{x} = x - 1$, where x is its displacement at time t .

- (i) Show that the velocity of the particle is given by $\dot{x} = 1 - x$. **3**
 - (ii) Find an expression for x as a function of t . **2**
 - (iii) Find the limiting position of the particle. **1**
- (c) Two players A and B play a series of games against each other to get a prize. In any game, either of the players is equally likely to win.

To begin with, the first player who wins a total of 5 games gets the prize.

- (i) Explain why the probability of player A getting the prize in exactly 7 games is $\binom{6}{4} \left(\frac{1}{2}\right)^7$. **1**
- (ii) Write an expression for the probability of player A getting the prize in at most 7 games. **1**
- (iii) Suppose now that the prize is given to the first player to win a total of $(n + 1)$ games, where n is a positive integer. **2**

By considering the probability that A gets the prize, prove that

$$\binom{n}{n}2^n + \binom{n+1}{n}2^{n-1} + \binom{n+2}{n}2^{n-2} + \cdots + \binom{2n}{n} = 2^{2n}.$$

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

2015 HSC Mathematics Extension 1 Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	A
2	C
3	B
4	C
5	A
6	D
7	B
8	B
9	D
10	C

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Finds the correct relationship between $\cos 2x$ and $\sin^2 x$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 & \int \sin^2 x \, dx \\
 &= \frac{1}{2} \int 1 - \cos 2x \, dx \\
 &= \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) + C
 \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the correct slopes in an appropriate formula, or equivalent merit	1

Sample answer:

$$m_1 = 2 \quad m_2 = -3$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{2 - (-3)}{1 + 2 \times (-3)} \right|$$

$$= \left| \frac{5}{-5} \right|$$

$$= 1$$

$$\therefore \text{acute angle} = 45^\circ \text{ or } \frac{\pi}{4}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	3
• Identifies both important values, or equivalent merit	2
• Identifies -3 as an important value, or equivalent merit	1

Sample answer:

$$\frac{4}{x+3} \geq 1$$

$$(x+3)^2 \times \frac{4}{(x+3)} \geq (x+3)^2 \times 1, (x \neq -3)$$

$$4(x+3) \geq (x+3)^2$$

$$0 \geq (x+3)^2 - 4(x+3)$$

$$0 \geq (x+3)(x+3-4)$$

$$0 \geq (x+3)(x-1)$$

$$\Rightarrow -3 < x \leq 1$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Finds A , or equivalent merit	1

Sample answer:

$$5\cos x - 12\sin x$$

$$A\cos(x+\alpha) = A\cos x \cos \alpha - A\sin x \sin \alpha$$

$$\therefore A\cos \alpha = 5$$

$$A\sin \alpha = 12$$

$$A = \sqrt{5^2 + 12^2}$$

$$= 13$$

$$\tan \alpha = \frac{12}{5}$$

$$\alpha = 1.176005207\dots$$

$$5\cos x - 12\sin x = 13\cos(x + 1.176005207\dots)$$

or

$$5\cos x - 12\sin x = 13\cos\left(x + \tan^{-1}\left(\frac{12}{5}\right)\right)$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Finds a correct expression for the integral, or equivalent merit	2
• Correctly attempts substitution, or equivalent merit	1

Sample answer:Let $u = 2x - 1$

$$\frac{du}{dx} = 2 \Rightarrow dx = \frac{1}{2} du$$

when $x = 1, u = 1$ $x = 2, u = 3$

$$\therefore \int_1^2 \frac{x}{(2x-1)^2} dx$$

$$= \int_1^3 \frac{1}{2} \left(\frac{u+1}{u^2} \right) \cdot \frac{1}{2} du$$

$$= \frac{1}{4} \int_1^3 \left(\frac{u+1}{u^2} \right) du$$

$$= \frac{1}{4} \int_1^3 \left(\frac{1}{u} + \frac{1}{u^2} \right) du$$

$$= \frac{1}{4} \left[\ln u - \frac{1}{u} \right]_1^3$$

$$= \frac{1}{4} \left[\left(\ln 3 - \frac{1}{3} \right) - \left(\ln 1 - \frac{1}{1} \right) \right]$$

$$= \frac{3\ln 3 + 2}{12}$$

Question 11 (f) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$P(x) = x^3 - kx^2 + 5x + 12$$

Since $x - 3$ is a factor, then $P(3) = 0$

$$\therefore (3)^3 - k(3)^2 + 5(3) + 12 = 0$$

$$27 - 9k + 15 + 12 = 0$$

$$9k = 54$$

$$k = 6$$

Question 11 (f) (ii)

Criteria	Marks
• Provides correct solution	2
• Identifies quadratic factor, or equivalent merit	1

Sample answer:

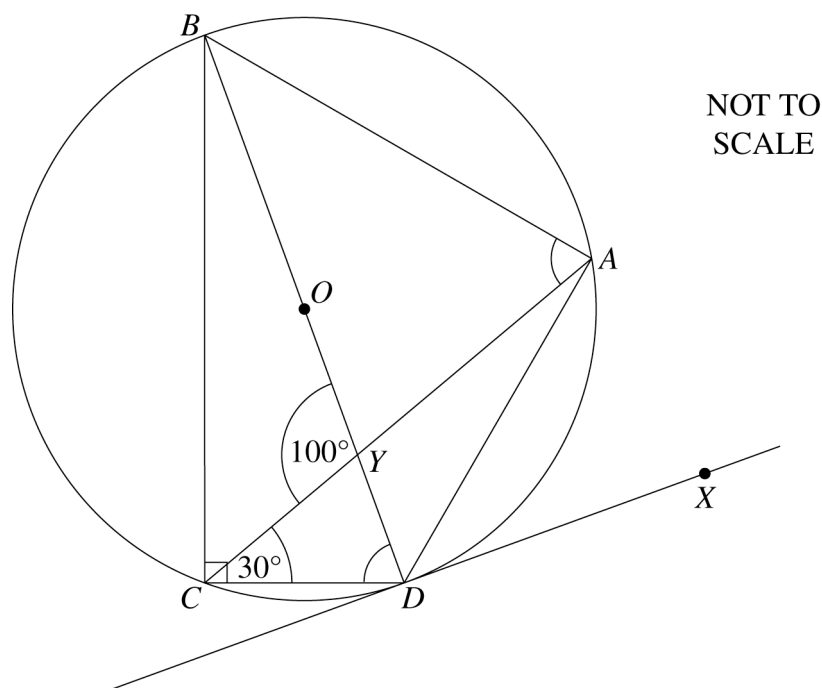
Using $k = 6$ $P(x) = x^3 - 6x^2 + 5x + 12$.

By long division

$$\begin{array}{r}
 x^2 - 3x - 4 \\
 x - 3 \overline{) x^3 - 6x^2 + 5x + 12} \\
 \underline{x^3 - 3x^2} \\
 -3x^2 + 5x \\
 \underline{-3x^2 + 9x} \\
 -4x + 12 \\
 \underline{-4x + 12} \\
 0
 \end{array}$$

$$\begin{aligned}
 \therefore P(x) &= (x - 3)(x^2 - 3x - 4) \\
 &= (x - 3)(x - 4)(x + 1)
 \end{aligned}$$

$\therefore$ zeros are 3, 4, -1

Question 12 (a)**Question 12 (a) (i)**

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}\angle ACB + \angle ACD &= 90^\circ && \text{(Angle in a semi-circle is a right angle)} \\ \angle ACB + 30^\circ &= 90^\circ \\ \therefore \angle ACB &= 60^\circ\end{aligned}$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}\angle ADX &= \angle ACD && \text{(angle between tangent and chord = angle in the alternate segment)} \\ &= 30^\circ\end{aligned}$$

Question 12 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Makes some progress towards correct solution	1

Sample answer:

$$\angle CDY + \angle YCD = 100^\circ \quad (\angle BYC, \text{ exterior } \angle \text{ of } \triangle CDY)$$

$$\therefore \angle CDY = 70^\circ$$

$$\angle CAB = \angle CDB \quad (\text{Angles standing on same chord at the circumference are equal})$$

$$\therefore \angle CAB = 70^\circ$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$PQ : (p + q)x - 2y - 2apq = 0$$

At the focus $x = 0$, $y = a$

$$\Rightarrow (p + q)0 - 2(a) - 2apq = 0$$

$$\therefore -2a = 2apq$$

$$\Rightarrow pq = -1$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct coordinates	2
• Identifies the value of p , or equivalent merit	1

Sample answer:

$$P(8a, 16a), \therefore 8a = 2ap$$

$$p = 4$$

$$q = -\frac{1}{p} \quad (\text{from (i)})$$

$$\therefore q = -\frac{1}{4}$$

$$\begin{aligned} \therefore Q(2aq, aq^2) &= \left(2a\left(-\frac{1}{4}\right), a\left(-\frac{1}{4}\right)^2 \right) \\ &= \left(-\frac{a}{2}, \frac{a}{16} \right) \end{aligned}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct solution	1

*Sample answer:*In $\triangle MOA$,

$$\tan 15^\circ = \frac{h}{OA}$$

$$\cot 15^\circ = \frac{OA}{h}$$

$$OA = h \cot 15^\circ$$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Uses Pythagoras' theorem, or equivalent merit	1

*Sample answer:*In $\triangle MOB$,

$$\tan 13^\circ = \frac{h}{OB}$$

$$\cot 13^\circ = \frac{OB}{h}$$

$$OB = h \cot 13^\circ$$

In $\triangle OAB$,

$$OB^2 = OA^2 + AB^2$$

$$h^2 \cot^2 13^\circ = h^2 \cot^2 15^\circ + 2000^2$$

$$h^2 (\cot^2 13^\circ - \cot^2 15^\circ) = 2000^2$$

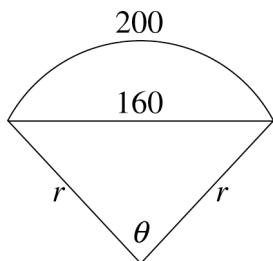
$$h^2 = \frac{2000^2}{\cot^2 13^\circ - \cot^2 15^\circ}$$

$$= 827561.0914\dots$$

$$h = 910$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:Using cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$

$$160^2 = r^2 + r^2 - 2r.r.\cos\theta$$

$$160^2 = 2r^2 - 2r^2 \cos\theta$$

$$\therefore 160^2 = 2r^2(1 - \cos\theta)$$

Question 12 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Observes $200 = r\theta$, or equivalent merit	1

*Sample answer:*Using $\ell = r\theta$

$$200 = r\theta$$

$$r = \frac{200}{\theta}$$

$$160^2 = 2\left(\frac{200}{\theta}\right)^2(1 - \cos\theta)$$

$$160^2\theta^2 = 2(200)^2(1 - \cos\theta)$$

$$8\theta^2 = 25 - 25\cos\theta$$

$$\therefore 8\theta^2 + 25\cos\theta - 25 = 0$$

Question 12 (d) (iii)

Criteria	Marks
• Provides correct solution	2
• Finds derivative of $8\theta^2 + 25\cos\theta - 25$, or equivalent merit	1

Sample answer:

$$\theta_1 = \pi \quad \text{Let } f(\theta) = 8\theta^2 + 25\cos\theta - 25$$

$$f'(\theta) = 16\theta - 25\sin\theta$$

$$\theta_2 = \theta_1 - \frac{f(\theta_1)}{f'(\theta_1)}$$

$$= \pi - \frac{8\pi^2 + 25\cos\pi - 25}{16(\pi) - 25\sin\pi}$$

$$= \pi - \frac{8\pi^2 - 50}{16\pi}$$

$$= 2.565514\dots$$

$$= 2.57 \text{ (two decimal places)}$$

Question 13 (a) (i)

Criteria	Marks
• Provides correct values	1

Sample answer:

$$x = 3 \text{ or } x = 7$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Max speed when $v^2 = 11$

$$\therefore \text{Max speed} = \sqrt{11} \text{ m s}^{-1}$$

Question 13 (a) (iii)

Criteria	Marks
• Provides correct solution	3
• Deduces the values of two of a , c , n , or equivalent merit	2
• Writes down the value of c , or equivalent merit	1

Sample answer:

$$c = 5 \quad (\text{particle oscillates about } x = 5)$$

$$a = 2 \quad (\text{amplitude is } 5 - 3 \text{ or } 7 - 5)$$

$$v^2 = n^2(2^2 - (x - 5)^2)$$

$$v^2 = 11 \text{ when } x = 5$$

$$\therefore 11 = 4n^2$$

$$\Rightarrow n^2 = \frac{11}{4}$$

$$\Rightarrow n = \frac{\sqrt{11}}{2}$$

$$(n > 0)$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Identifies $\binom{18}{2}$ or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \left(2x + \frac{1}{3x}\right)^{18} &= a_0x^{18} + a_1x^{16} + a_2x^{14} + \dots \\
 &= \binom{18}{0}(2x)^{18}\left(\frac{1}{3x}\right)^0 + \binom{18}{1}(2x)^{17}\left(\frac{1}{3x}\right)^1 + \binom{18}{2}(2x)^{16}\left(\frac{1}{3x}\right)^2 + \dots \\
 \therefore a_2 &= \binom{18}{2}2^{16}\frac{1}{3^2} \\
 &= 1\,114\,112
 \end{aligned}$$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Identifies correct term, or equivalent merit	1

Sample answer:

$$T_{k+1} = \binom{18}{k}(2x)^{18-k}\left(\frac{1}{3x}\right)^k$$

For constant term, $x^{18-k} \cdot x^{-k} = x^0$

$$\Rightarrow 18 - 2k = 0$$

$$\therefore k = 9$$

$\therefore$ constant term/term independent of x is $\binom{18}{9} \cdot 2^9 \cdot \left(\frac{1}{3}\right)^9$

Question 13 (c)

Criteria	Marks
• Provides correct proof	3
• Verifies initial case and attempts to use inductive assumption, or equivalent merit	2
• Verifies the identity for the initial case, or equivalent merit	1

Sample answer:

$$\begin{aligned}\text{When } n = 1, \text{ RHS} &= 1 - \frac{1}{(1+1)!} \\ &= 1 - \frac{1}{2!} = \frac{1}{2} = \text{LHS}\end{aligned}$$

Statement true for $n = 1$.Assume the result is true for $n = k$

$$\text{ie assume } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!} \text{ to be true.}$$

Prove true for $n = k + 1$

$$\text{ie } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+1+1)!} = 1 - \frac{1}{(k+1+1)!}$$

$$\begin{aligned}\text{LHS} &= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad (\text{from assumption}) \\ &= 1 - \frac{(k+2) - (k+1)}{(k+2)!} \\ &= 1 - \frac{1}{(k+2)!} = \text{RHS}\end{aligned}$$

Hence the result is proven by mathematical induction.

Question 13 (d) (i)

Criteria	Marks
• Provides correct proof	2
• Correctly differentiates $\cos^{-1}x$, or equivalent merit	1

Sample answer:

$$f(x) = \cos^{-1}(x) + \cos^{-1}(-x)$$

$$f'(x) = \frac{-1}{\sqrt{1-x^2}} + -1 \cdot \frac{-1}{\sqrt{1-(-x)^2}} = \frac{-1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}$$

$$= 0$$

$$\therefore f'(x) = 0$$

$$\therefore f(x) \text{ is constant}$$

Question 13 (d) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:Substitute any suitable value, eg $x = 0$

$$f(0) = \cos^{-1}(0) + \cos^{-1}(0)$$

$$= \frac{\pi}{2} + \frac{\pi}{2}$$

$$\therefore f(x) = \pi$$

$$\therefore \cos^{-1}(x) + \cos^{-1}(-x) = \pi$$

$$\text{ie } \cos^{-1}(-x) = \pi - \cos^{-1}(x)$$

Question 14 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Finds correct time when $y = 0$, or equivalent merit	1

Sample answer:When $y = 0$

$$0 = Vt \sin \theta - \frac{1}{2}gt^2$$

$$0 = t \left(V \sin \theta - \frac{1}{2}gt \right)$$

$$t = 0 \quad \text{or} \quad t = \frac{2V \sin \theta}{g}$$

Range: $x = Vt \cos \theta$ when $t = \frac{2V \sin \theta}{g}$

$$= V \left(\frac{2V \sin \theta}{g} \right) \cos \theta$$

$$= \frac{2V^2 \sin \theta \cos \theta}{g}$$

$$= \frac{V^2 \sin 2\theta}{g} \quad (\sin 2\theta = 2 \sin \theta \cos \theta)$$

Question 14 (a) (ii)

Criteria	Marks
• Provides a correct angle to the horizontal	2
• Provides $\dot{x}$ and $\dot{y}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{dx}{dt} &= V \cos \theta \\ \frac{dy}{dt} &= V \sin \theta - gt \\ \therefore \frac{dy}{dx} &= \frac{V \sin \theta - gt}{V \cos \theta} \\ \text{When } \theta &= \frac{\pi}{3} \text{ and } t = \frac{2V}{\sqrt{3}g} \\ \frac{dy}{dx} &= \frac{V \sin \frac{\pi}{3} - g \left(\frac{2V}{\sqrt{3}g} \right)}{V \cos \frac{\pi}{3}} \\ &= \frac{V \cdot \frac{\sqrt{3}}{2} - \frac{2V}{\sqrt{3}}}{V \cdot \frac{1}{2}} \\ \frac{dy}{dx} &= -\frac{1}{\sqrt{3}}\end{aligned}$$

Let α be the angle between the curve and the positive direction of the x -axis at that time.

$$\therefore \tan \alpha = -\frac{1}{\sqrt{3}}$$

$$\Rightarrow \alpha = \frac{5\pi}{6} \text{ from positive } x\text{-axis}$$

Question 14 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:Since α is obtuse, the projectile is travelling downwards (or is descending).

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	3
• Writes v^2 as a perfect square, or equivalent merit	2
• Attempts to use $\frac{d\left(\frac{1}{2}v^2\right)}{dx}$, or equivalent merit	1

Sample answer:

$$\ddot{x} = x - 1 \quad t = 0, x = 0, v = 1$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = x - 1$$

$$\frac{1}{2}v^2 = \frac{1}{2}x^2 - x + C_1$$

$$v^2 = x^2 - 2x + 2C_1$$

$$\text{At } x = 0, v = 1, 1 = 0 - 0 + 2C_1$$

$$2C_1 = 1$$

$$\therefore v^2 = x^2 - 2x + 1$$

$$v^2 = (x - 1)^2$$

$$\therefore v = \pm(x - 1)$$

Now $v = 1$ when $x = 0$ so

$$1 = -(0 - 1)$$

ie take negative case

$$\therefore v = -(x - 1)$$

$$\therefore v = 1 - x$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct expression	2
• Attempts to integrate a correct expression for $\frac{dt}{dx}$, or equivalent merit	1

Sample answer:

$$v = 1 - x$$

$$\therefore \frac{dx}{dt} = 1 - x$$

$$\therefore \frac{dt}{dx} = \frac{1}{1 - x}$$

$$t = -\ln(1 - x) + C_2$$

When $t = 0$ $x = 0$

$$\therefore C_2 = 0$$

$$\therefore t = -\ln(1 - x)$$

$$-t = \ln(1 - x)$$

$$\therefore e^{-t} = 1 - x$$

$$\therefore x = 1 - e^{-t}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$x = 1 - e^{-t}$$

As $t \rightarrow \infty$, $e^{-t} \rightarrow 0$

$$\Rightarrow x \rightarrow 1 - 0$$

$$= 1$$

 $\therefore$ limiting position is $x = 1$

Question 14 (c) (i)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

To get a prize in 7 games, A must win 4 of the first 6 games and then win the 7th.

$$\begin{aligned}\therefore \text{Prob} &= \binom{6}{4} \left(\frac{1}{2}\right)^6 \times \frac{1}{2} \\ &= \binom{6}{4} \left(\frac{1}{2}\right)^7\end{aligned}$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct expression	1

Sample answer:

A can get a prize by winning in 5 games or 6 games or 7 games.

$$\begin{aligned}\text{Prob} &= \binom{4}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right) + \binom{5}{4} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right) + \binom{6}{4} \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right) \\ &= \binom{4}{4} \left(\frac{1}{2}\right)^5 + \binom{5}{4} \left(\frac{1}{2}\right)^6 + \binom{6}{4} \left(\frac{1}{2}\right)^7\end{aligned}$$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct proof	2
• Recognises that the probability that A wins the prize is $\frac{1}{2}$, or equivalent merit	1

Sample answer:

A can win in $n + 1$ games or $(n + 2)$ games or $(n + 3)$ games ... or $2n$ games.

$$\text{Prob} = \binom{n}{n} \left(\frac{1}{2}\right)^{n+1} + \binom{n+1}{n} \left(\frac{1}{2}\right)^{n+2} + \binom{n+2}{n} \left(\frac{1}{2}\right)^{n+3} + \dots + \binom{2n}{n} \left(\frac{1}{2}\right)^{2n+1}$$

But probability that A eventually wins is $\frac{1}{2}$

Equating these expressions and multiplying by 2^{2n+1} gives

$$\binom{n}{n} 2^n + \binom{n+1}{n} 2^{n-1} + \binom{n+2}{n} 2^{n-2} + \dots + \binom{2n}{n}$$

$$= \frac{1}{2} \times 2^{2n+1}$$

$$= 2^{2n}$$

2015 HSC Mathematics Extension 1

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	16.2E	PE3
2	1	14.2E	HE3
3	1	2.9E	PE3
4	1	18.1E	PE3
5	1	10.5E	H5, HE7
6	1	15.2E	HE4
7	1	15.5E	HE4
8	1	8.1, 8.2, 13.4, 13.5E	HE7
9	1	14.4E	HE3
10	1	13.7	H5

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	13.6E	HE6
11 (b)	2	6.6E	P4
11 (c)	3	1.4E	PE2, PE3
11 (d)	2	5.9E	P4
11 (e)	3	11.5E	HE6
11 (f) (i)	1	16.2E	PE3
11 (f) (ii)	2	16.2E	PE2, PE3
12 (a) (i)	1	2.8, 2.10	PE3
12 (a) (ii)	1	2.8, 2.10	PE3
12 (a) (iii)	2	2.8, 2.10	PE2, PE3
12 (b) (i)	1	9.6E	PE3, PE4
12 (b) (ii)	2	9.6E	PE3, PE4
12 (c) (i)	1	5.6E	PE6
12 (c) (ii)	2	5.6E	H5
12 (d) (i)	1	5.5	P4
12 (d) (ii)	2	13.1	H5
12 (d) (iii)	2	16.4E	HE7
13 (a) (i)	1	14.4E	HE3
13 (a) (ii)	1	14.4E	HE3
13 (a) (iii)	3	14.4E	HE3
13 (b) (i)	2	17.3E	HE7
13 (b) (ii)	2	17.3E	HE7
13 (c)	3	7.4E	HE2
13 (d) (i)	2	10.1, 10.2, 15.5E	HE4

Question	Marks	Content	Syllabus outcomes
13 (d) (ii)	1	15.4E	HE4
14 (a) (i)	2	14.3E	HE3
14 (a) (ii)	2	14.3E	HE3
14 (a) (iii)	1	14.3E	HE3
14 (b) (i)	3	14.3E	HE5
14 (b) (ii)	2	14.3E	HE5
14 (b) (iii)	1	14.2E	H3, H4, HE7
14 (c) (i)	1	17.1E, 17.3E, 18.1E, 18.2E	HE3
14 (c) (ii)	1	17.1E, 17.3E, 18.1E, 18.2E	HE3
14 (c) (iii)	2	17.1E, 17.3E, 18.1E, 18.2E	HE3